Prifysgol Glyndŵr Wrecsam | Wrexham Glyndŵr University

COM 736

Database Systems and Data Analytics (Week 9)

Big Data Fundamentals and Architecture

Big Data

What is Big Data ?

- Large Volume of Data
- Varied Source and format
- Velocity

Why Big Data ?

- Gain critical and useful insights
- Machine Learning
- Simpler and error free analysis.

Why now?

- modern technologies along with new data processing frame works and platforms
- Increased scalability and flexibility of the processing architecture
- Automated data processing capability

Big Data

- Each organisation will have its own data requirements for big data processing
  - Weather Data
    - Governments agencies, scientific organisations, and consumers like farmers, Television channels or radio channels
  - Clinical Trials Data
    - Pharmaceutical companies
    - Personalised Vaccines?
  - Epidemiology
    - Shaping the policy decisions as well as evidence-based practice
    - Disease prevention and control strategies
    - Cornerstone of public health

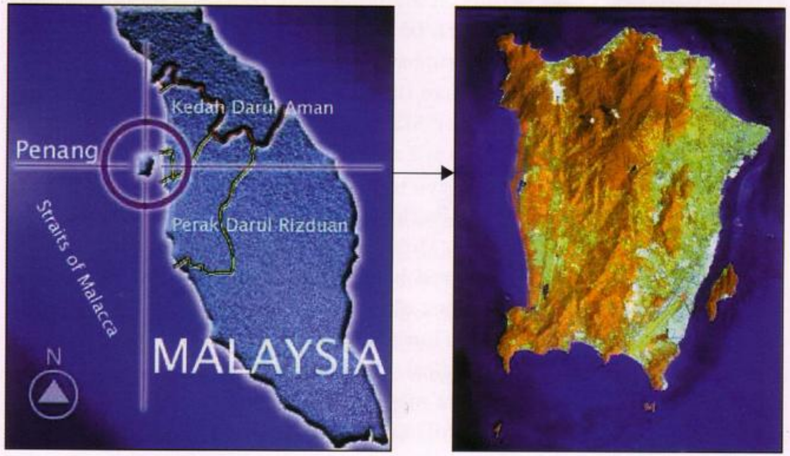
Example

- Fast Food Restaurant [1]
  - A fast-food restaurant chain wants to know the correlation between its sales and consumer traffic based on weather conditions.
  - Historic pattern and current pattern needs to be integrated and analysed, social media sharing of consumer perspectives about shopping experience and were weather is mentioned is a key factor.

[1] K. Krishnan, Data Warehouse In The Age of Big Data, Elsevier, 2013.

Example Earth Imaging analysis

- - using algorithm over earth imagine dataset
- Highlighting the most important (spectrally-observable) phenomena of crop development in a way that allows discrimination of specific crops.



(1a)
Figure 1a. Penang Island on the Malaysian Map (Source: Penang Map and Interesting Locations, 2002)

(1b)
Figure 1b. Penang Island from Landsat-5 TM image acquired on 3 March 1998

Kasturi Devi Kanniah and Phoey Lee,Generation of Tasseled Cap Transformation Coeffiecients for the use of TiungSAT-1 Multi Spectral Earth Imaging System (MSEIS) Data

Sample Data set for Analysis

Data	Features	Source	Complexity
Weather	Structured and Semis Structured Available for all latitude and longitude codes Has weather persons editorial commentary and user shared data	Governmental Agencies Public News Channels Social Media	Metadata Geo Coding Language Image and Video formats
Consumer Sentiment	Voice Text Images Videos Blogs, Forums, and other Internet based comments and likes	Call Centre Social Media Campaign Database Customer Resource Management (CRM)	Metadata Context Language Formats
Product Competition	Corporate Product Data Available in structured doormats from data federators Available unstructured from social media, forum and Internet	In-House Third Party Social Media Internal Research	Hierarchies Metadata Data Quality Context
Location	Structured Unstructured	Internal Social Media Third Party Surveys	Metadata Data Quality Formats: Images and Videos
Campaign	Structured	Internal	Multiple campaigns at any given time across geographies

Sample Social Media Analytics

- To understand how intensive is profanity usage in social media platform to visualise the profanity classes in text communication among users from different countries – *Phoey Lee Teh & CB Cheng 2022, Profane Words on Twitter: an analysis of the use of swear words in different countries.*
- To explores a set of topic modelling techniques to assist policymakers and environment communities in understanding public opinions about the issues related to plastic pollution by analysing social media data, 274,404 tweets were collected from Twitter that are related to plastic pollution - *Teh, P.L.; Piao, S.; Almansour, M.; Ong, H.F.; Ahad, A. Analysis of Popular Social Media Topics Regarding Plastic Pollution. Sustainability 2022, 14, 1709.*

Sample Social Media Analytics

- to analyze tweets coming from casual Twitter users and twitter accounts representing the veganism society and industry, We collected n = 50,634 tweets and analyzed n = 25,639 processed tweets. - *Phoey Lee Teh & Wei Li Yap (2021), GoVegan: Exploring Motives and Opinions from Tweets,WorldCIST 2021, AISC 1366, pp. 3–12, 2021. https://doi.org/10.1007/978-3-030-72651-5_1*

Sample Social Media Analytics

- examined the factors associated with Airbnb utilization from various platforms, but not exclusively from Twitter. A total of 21,097 tweets was collected in a period of two months, and the tweets were qualitatively analyzed with the help of text analysis tools to verify the discourse of discussion. - *Phoey Lee Teh, YCLow, PBOoi, Viewing Airbnb from Twitter: factors associated with users' utilization, . In (iiWAS2020). ACM, New York, NY, USA, 9 pages. https://doi.org/3428757.3429112*

Integration Analysis

- Large volume of data available from different sources and in varied format
- Few question that could be answered from the wealth of data
  - What sales occurred across the US for a given day/week/month/quarter/year, and under various weather conditions?
  - Did people prefer drive-thru in extreme weather conditions irrespective of the geography?
  - Did restaurants along the highway get more traffic in drive-thru during regular versus abnormal weather?
  - Did service interruptions occur due to weather?
  - What is the tendency for business impact in abnormal weather?
  - Does the restaurant need to staff more in different weather conditions? What is the budget impact in such situations?
  - Does coffee sell more than burgers in the winter in Boston, and during the same time, do more cold beverages sell in Orlando?
  - What are the customer expectations of pricing? Do they give feedback by phone, email or social media?
  - What is the competition comparison by customers in social media?

Integration and Analysis

Better understanding of

- Customers
- Markets
- Products
- Vendors/Suppliers
- Staff Management
- Campaign
- Location Management

Better insight into

- What drives the business
- Impact / Power of the weather
- The business and the consumer.

Value of Data

Data Growth is characterised by

- Volume (amount of Data),
- Velocity (speed of data in and out),
- Variety (range of data type and sources).

Business model transformation

- Service oriented rather than product oriented.
- Success is measured by the customer satisfaction not by the usefulness of the product (?).
- Huge amount of supporting data is generated through various mediums. (Volume)
- Globalisation: key trend, changed the format of the data. (Variety)
- Personalisation of services, generating data with every click. (Velocity)

Data Volume

Machine Data

- Steady Patterns of numbers and Text, occurs in a rapid fire fashion
- Sensor Data
- Application Log
- Click Stream Data

External data

- Data feed from external parties
- Weather data
- Emails

Contracts

- Human resources
- Legal
- Vendor
- Supplier
- Customer etc.

Data Volume

Geo-Spatial Data

- Mobile devices use GPS for navigation and personalisation.
- GPS added to the camera’s and camcorders to set the location of the picture

Social media

- Facebook
- Twitter
- Instagram

Companies must make use of the available data

- Different format
- Large volume
- Speed etc.

Data Velocity

Speed and direction of motion of an (physical) object.

Data needs to be processed in a continuous stream, not as a batch

For example websites tracking the customer clicks and navigation and provide personalised browsing experience

Millions of clicks every second resulting in large amount of data

The speed of the data generated here demonstrates Data Velocity

Data Velocity

Sensor data

- Data comes from variety of sensors, GPS, tyre pressure systems, temperature sensors, mobile devices

Mobile networks

- Sharing data and pictures through mobile device, volume and velocity

Social Media

- Twitter, Facebook, YouTube, Flickr, data at different velocities, multiple format

Data velocity

- Along with the data volume will be tricky for RDBMS, creating silos of solutions to process the data type means scalability will be an issue.

Data Variety

Big Data comes in different data formats

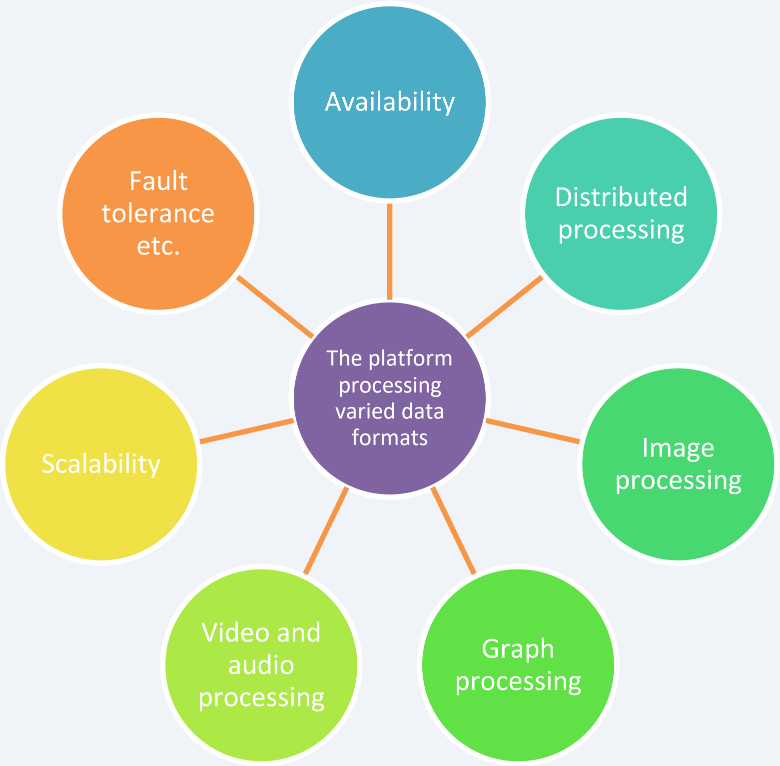
Every form of data is important

Appropriate metadata is very critical.

Absence of or partial metadata means processing delays

Emails, Tweets, Social Media, Sensor Data, Video, Audio….

Data Variety



Big Data

More Vs

3 Vs, Volume, Velocity and Variety

Can be extended with

Vagueness

Ambiguity, lack of metadata

In a photograph or in a graph M and F could mean Male/Female, or Monday/Friday

Virality

How quickly the data is shared between people-to-people through a peer network.

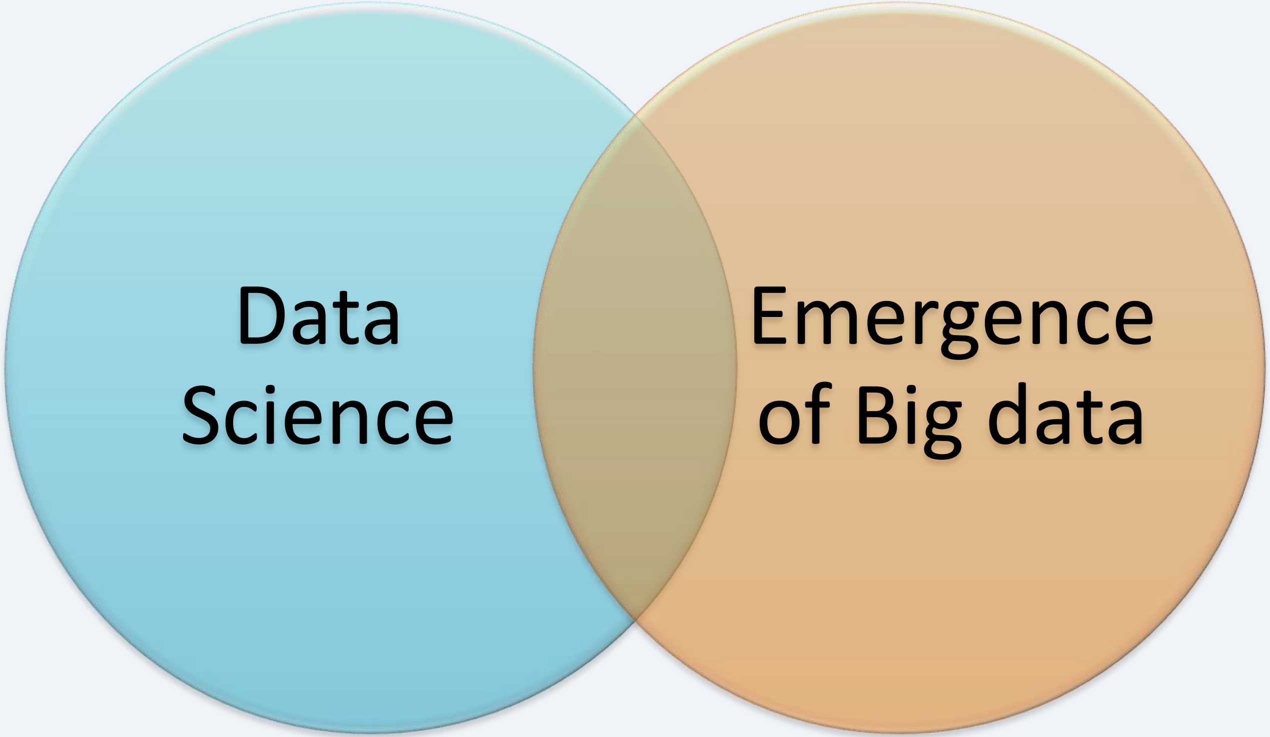
Understanding the context is important…

Viscosity

Measures the resistance (slow down) to flow in the volume of the data.

Resistance can manifest in data flows, business rules or even be the limitation of the technology.

Summary



Next

Big Data Technologies

An overview various components